





AI-Based Smart Expense Management System with Automated Categorization and Budget Recommendations

Abstract

Managing personal finances effectively has become increasingly important in today's digital era due to the growing number of digital transactions and daily expenses. The proposed AI-Based Smart Expense Management System with Automated Categorization and Budget Recommendations automatically captures expense details from payment notifications, categorizes transactions using Artificial Intelligence, analyzes spending patterns, predicts future expenses using Linear Regression, detects unusual spending through Isolation Forest, and provides personalized budget recommendations. The system extracts transaction details such as merchant name and amount, stores them securely, and presents users with real-time dashboards and financial insights. The application is developed using Python, Flask, SQLite, Machine Learning techniques, HTML, CSS, and JavaScript to provide a secure, user-friendly, and responsive interface. The proposed system minimizes manual effort, improves financial awareness, and enables users to make informed financial decisions through intelligent automation.

Keywords: Artificial Intelligence; Smart Expense Management; Automated Categorization; Budget Recommendations; Notification Reader; Linear Regression; Naive Bayes; Isolation Forest.

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